LAB 02

Data-Validation Stage: Cross-Check Pipeline Bars against TradingView

Item	Detail
Topic	Topic 1: Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents
Pipeline stage	Data validation
TSC reference	A3, K5, K9
Duration	20 min
Folder	labs_fullcode/Lab02-tradingview-charts/
Files	Lab02-tradingview-charts/validate_stage.py, Lab02-tradingview-charts/lcc_data_check.pine, Lab02-tradingview-charts/tradingview_values.csv, pipeline/validate.py, pipeline/reference/AAPL_1Day_raw.csv, pipeline/reference/D05.SI_1Day_raw.csv, Lab02-tradingview-charts/solution/validate.py
Prompts	Lab02_Prompts.pdf in this folder (step -> prompt, ready to copy)
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital's traders watch DBS Group (SGX:D05) and Apple (NASDAQ:AAPL) on TradingView, while the pipeline trains and trades on data from an API. Before any model is built, the desk must prove that the pipeline's bars agree with what traders see on the chart - and understand why adjusted research prices differ from the unadjusted prices on the screen and on an order ticket.

Objective

Read daily bars for SGX:D05 and NASDAQ:AAPL from the TradingView Data Window, record them, and build a validation stage that compares TradingView with the pipeline's raw and dividend-adjusted bars and writes a validation report.

Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

How to Run the Python Scripts

Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.

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Step	Command / file
Open a terminal in the labs_fullcode folder	cd labs_fullcode
Create and activate a virtual environment (once)	python3 -m venv .venv source .venv/bin/activate (Windows: .venv\Scripts\activate)
Install the libraries (once)	pip install -r requirements.txt
Optional: your Alpaca paper keys (leave blank to run OFFLINE)	cp .env.example .env (Windows: copy .env.example .env) then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env
Run Lab 02	python3 Lab02-tradingview-charts/validate_stage.py (same as: python -m pipeline validate)
The complete code	pipeline/validate.py (reference copy: Lab02-tradingview-charts/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

Lab Steps - In Python with Your AI Assistant

AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab02-tradingview-charts/Lab02_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.

Step 1: Add the data-check panel to SGX:D05 (A3, K9)

Log in to tradingview.com and open a chart of SGX:D05, timeframe 1D, chart type Candles. Open the Pine Editor, paste the contents of Lab02-tradingview-charts/lcc_data_check.pine, Save it as 'LCC Data Check' and click Add to chart. In its Settings set 'Bar to validate' to 31 December 2025. Open the Data Window (right toolbar) and hover over that bar.

AI prompt - paste into your AI coding assistant
Explain labs_fullcode/Lab02-tradingview-charts/lcc_data_check.pine for a finance professional: what each input does, how the script finds the bar to validate, and which values in its table I should copy into tradingview_values.csv. Do not change the file.

Check: The orange-shaded bar is 2025-12-31 and the panel shows about open 56.50, high 56.60, low 56.11, close 56.36, volume about 2.05 million (SGD, default unadjusted view). Free plans may show SGX data delayed.

Step 2: Repeat for NASDAQ:AAPL and switch the adjustment (A3, K9)

Open NASDAQ:AAPL on 1D with the same indicator and read 31 December 2025. Then open Chart settings > Symbol and tick 'Adjust data for dividends'; read the bar again and untick it.

Check: Default view: open 273.06, high 273.68, low 271.75, close 271.86, volume about 27.29 million. With 'Adjust data for dividends' ticked the close drops to about 271.12.

Step 3: Record what TradingView shows (A3)

Open Lab02-tradingview-charts/tradingview_values.csv. Overwrite the two example rows with your own readings (tradingview_symbol, date, open, high, low, close, volume, setting = default or adjusted).

Check: Two rows: NASDAQ:AAPL and SGX:D05, both dated 2025-12-31, setting default.

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Step 4: Build the validation stage (A3, K5, K9)

Ask your assistant for pipeline/validate.py. The reference unadjusted files are already in pipeline/reference/ (split-adjusted, dividend-unadjusted bars with an adj_close column, the view TradingView shows by default).

AI prompt - paste into your AI coding assistant
Create labs_fullcode/pipeline/validate.py. reference_bars(symbol) reads pipeline/reference/<SYMBOL>_1Day_raw.csv (columns timestamp, open, high, low, close, adj_close, volume, dividend, split). adjusted_bars(symbol, raw) returns the research bars from common.data.get_bars for the same dates and, if no file exists (FileNotFoundError, e.g. D05.SI), raw open/high/low/close multiplied by adj_close / close. compare(tv_row, bar) returns the percentage difference per field and match=True if every price is within 0.1% and volume within 10%. run(date='2025-12-31', tv_file=None) loops over config universe.watch, prints for each symbol the RAW row, the ADJUSTED row, your TradingView row from Lab02-tradingview-charts/tradingview_values.csv (lines starting with # are comments) and the differences, the dividend-adjustment factor (adjusted close / raw close), change vs prior close, 52-week high/low and 20-day average volume, and a verdict: 'PASS - TradingView matches RAW', 'PASS - TradingView matches ADJUSTED' or 'FAIL'. Save _outputs/Lab02_validation_report.csv and a chart Lab02_validation.png (raw vs adjusted close in 2025, one panel per symbol) with common.plotting.save. main(argv) supports --date and --tv. This stage only reads data - it must never call an order API. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: pipeline/validate.py exists and python -m pipeline validate --help lists --date and --tv.

Step 5: Run the validation stage (A3, K9)

Run: python -m pipeline validate --date 2025-12-31 (same as: python3 Lab02-tradingview-charts/validate_stage.py --date 2025-12-31).

Check: AAPL RAW 273.06 / 273.68 / 271.75 / 271.86 vs ADJUSTED 272.32 / 272.94 / 271.01 / 271.12, factor 0.9973, diff vs adjusted +0.272%. D05 RAW 56.50 / 56.60 / 56.11 / 56.36 vs ADJUSTED 54.35 / 54.45 / 53.98 / 54.22, factor 0.9620, diff +3.950%. Both 'PASS - TradingView matches RAW (dividend-unadjusted) bars'; RESULT: 2/2; _outputs/Lab02_validation_report.csv and Lab02_validation.png.

Step 6: Understand adjusted versus unadjusted prices (K9)

Open pipeline/reference/D05.SI_1Day_raw.csv and find the dividend and split columns (for example the 1.1 bonus-share split on 2024-04-22). Decide which series a backtest must use and which price an order ticket uses.

AI prompt - paste into your AI coding assistant
Using the dividend and split columns in labs_fullcode/pipeline/reference/D05.SI_1Day_raw.csv, explain why the pipeline's adjusted D05 close for 2025-12-31 is about 3.95% below TradingView's close, why backtests must use adjusted prices while orders are priced in raw prices, and what error a trading program makes if it mixes the two. Answer in five bullet points; do not change any file.

Check: D05's factor 0.9620 reflects dividends paid after the bar (back-adjusted into history); backtests use adjusted returns, orders use the raw market price.

Step 7: Make the validation fail on purpose (A3)

In tradingview_values.csv change the AAPL row to the dividend-adjusted values open 272.32, high 272.94, low 271.01, close 271.12, setting adjusted, and rerun the stage. Next change only the close to 280.00 and rerun. Finally restore your real reading.

Check: With 271.12 AAPL shows 'PASS - TradingView matches ADJUSTED bars'; with 280.00 it shows 'FAIL - TradingView matches neither series' and RESULT: 1/2.

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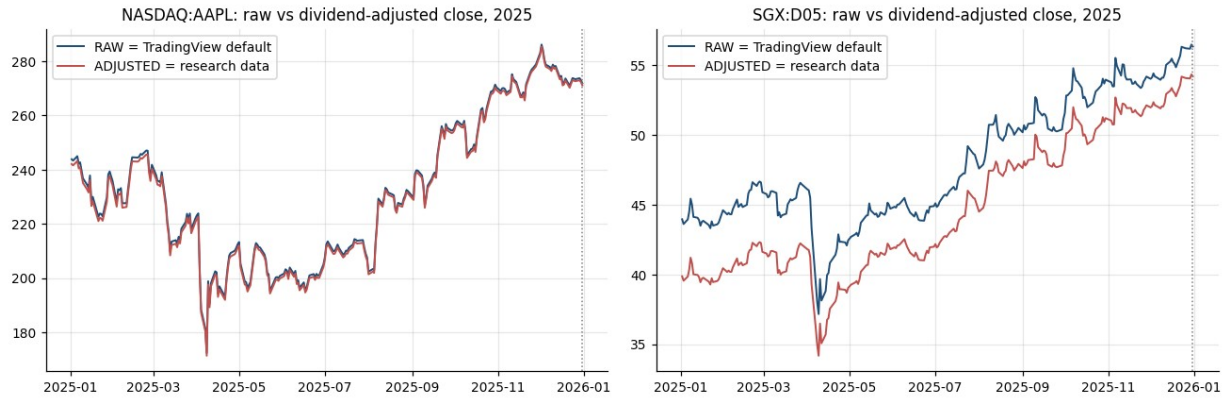
Step 8: Identify other forms of market data (K9)

On TradingView open the News, Financials and economic calendar views for SGX:D05. Note whether the SGX data are real-time or delayed on your plan and why the pipeline takes its US data from the Alpaca API rather than from a chart.

Check: You can name at least three data forms besides bars (news, fundamentals, economic calendar) and state whether your SGX data are delayed.

Expected Output (Full Code)

python -m pipeline validate --date 2025-12-31: AAPL RAW close 271.86 vs ADJUSTED 271.12 (factor 0.9973); SGX:D05 RAW close 56.36 vs ADJUSTED 54.22 (factor 0.9620); both match TradingView's default (unadjusted) view to the cent - 'PASS - TradingView matches RAW'; RESULT: 2/2 watch-list symbols validated; files _outputs/Lab02_validation_report.csv and _outputs/Lab02_validation.png.



Lab 2 output (labs_fullcode/_outputs/Lab02_validation.png), offline mode on the bundled data.

Guide: Sign Up for TradingView and Find the Tools You Will Use

The free TradingView plan is enough for Labs 2, 5 and 6. Do this before Lab 2.

1. Create the account

Go to tradingview.com and choose Sign up (top right). Register with email, Google or Apple, then confirm the activation email.

2. Choose the free plan

Keep the free plan when asked. It shows adverts and limits the number of indicators per chart and the alert features; webhook alerts (Lab 17) need a paid plan.

3. Open Supercharts

From Products choose Supercharts. Search a symbol such as NASDAQ:AAPL or SGX:D05 in the symbol box at the top left and choose the daily (1D) timeframe.

4. Set the time zone

Click the time zone at the bottom right of the chart and choose (UTC+8) Singapore, so bar times match the course examples.

5. Locate the four panels the labs use

Indicators (top toolbar), the Pine Editor and the Strategy Tester (tabs at the bottom), and Trade / the Trading Panel (top right) for paper trading.

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The free plan covers Labs 2, 5 and 6. Webhook alerts in Lab 17 need a paid plan - the lab includes a local simulator.

Debrief

Charting platforms are analytics tools (A3) and the API is the pipeline's data path (K5); a validation stage proves they agree before anything is automated. Knowing that adjusted and unadjusted prices are different forms of the same data (K9) prevents a classic error: training on adjusted history and then trading on raw quotes without converting.

Troubleshooting

Symptom	What to do
SGX:D05 shows delayed or no intraday data	Free plans show delayed SGX data; daily bars are enough for this lab.
Verdict FAIL for D05	Check the date, that the chart is 1D and that 'Adjust data for dividends' is unticked; enter the regular-session close, not an extended-hours price.
Data Window not visible	Open it from the right toolbar (Data Window icon) or press Alt+D.
Pine compile error	The first line must be //@version=6 and the whole file must be pasted.

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